Project Report: Semantic Segmentation Comparison

Oxford-IIIT Pet Dataset

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1 Problem Description

The objective of this project was to implement and compare two deep learning architectures for semantic segmentation on the Oxford-IIIT Pet Dataset. The task required pixel-level classification into three distinct classes:

1. **Pet:** The main body of the cat or dog.
2. **Background:** The surrounding environment.
3. **Boundary:** The outline or trimap area surrounding the pet.

2 Implementation Details

Two specific Convolutional Neural Network (CNN) architectures were implemented using the PyTorch framework:

2.1 Custom U-Net Architecture

A "from-scratch" U-Net was developed featuring a symmetric encoder-decoder structure.

- **Encoder:** Consists of convolutional layers and Max Pooling to extract high-level features.
- **Decoder:** Uses Transposed Convolutions to upsample the feature maps back to the original 128×128 resolution.
- **Skip Connections:** Critical connections that concatenate early encoder features with decoder features to preserve spatial detail and sharp boundaries.

2.2 FCN-ResNet50

As a baseline comparison, a Fully Convolutional Network (FCN) with a ResNet-50 backbone was utilized. This model represents a deeper architecture often used in industry-standard segmentation tasks.

3 Experimental Results

Training was conducted on an NVIDIA T4 GPU. The models were trained using **CrossEntropyLoss** and the **Adam** optimizer for a duration of 3 epochs.

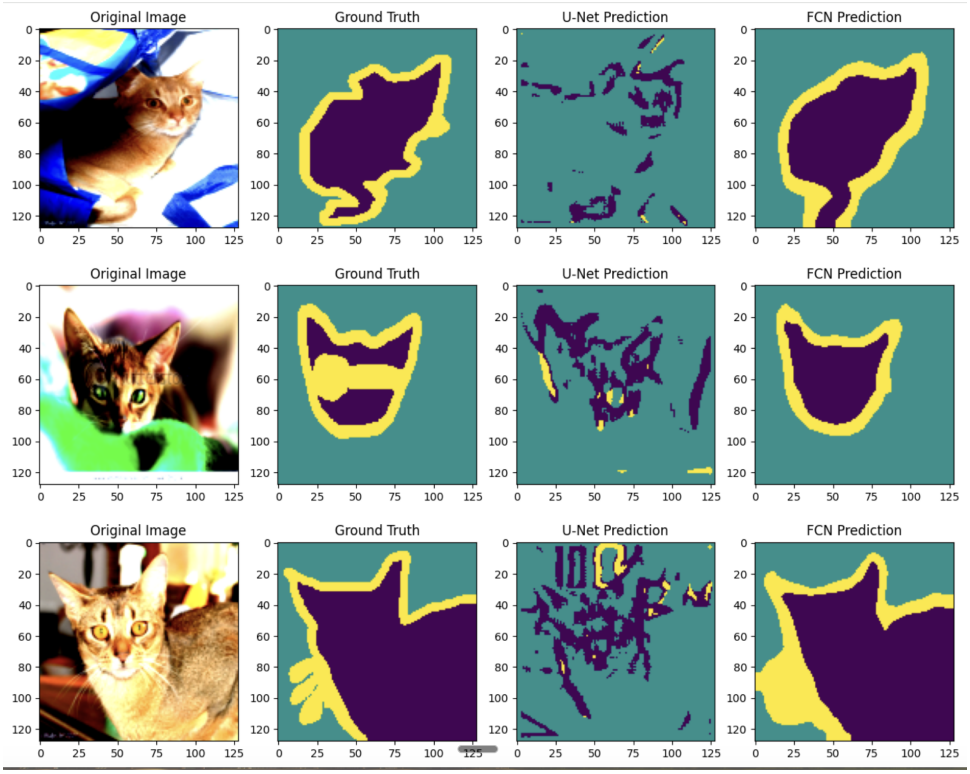

Figure 1: Qualitative comparison between Ground Truth, U-Net, and FCN predictions.

Metric	U-Net (From Scratch)	FCN-ResNet50
Accuracy	Lower (Fragmented)	High (Cohesive)
Edge Definition	Pixelated	Smooth
Training Setup	Random Initialization	Pre-trained Backbone

Table 1: Comparison of Model Performance after initial training epochs.

4 Failure Cases and Analysis

The primary failure case observed was the **fragmentation of the U-Net prediction masks**. While the FCN produced near-perfect masks, the U-Net results appeared as "scattered pixels" that vaguely resembled the pet's shape but lacked structural integrity.

Reasons for failure in U-Net:

- **Lack of Pre-training:** The FCN benefited from weights pre-trained on ImageNet, whereas the U-Net started with random weights.
- **Convergence Time:** Two epochs were insufficient for a randomly initialized U-Net to learn global geometric structures.
- **Depth:** The ResNet-50 backbone is significantly deeper than the implemented U-Net, allowing it to extract more complex features immediately.

5 Conclusion

While U-Net is architecturally superior for fine-detail segmentation due to skip connections, this project demonstrates that **Transfer Learning** (via FCN-ResNet50) provides a massive advantage when training data or time is limited.